DEPARTMENTOFELECTRONICSANDCOMMUNICATION ENGINEERING

AI Voice Assistant Using ESP32-S3 with Cloud AI and the XiaoZhi AI Framework

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ESP32-S3 AI Voice Assistant

ABSTRACT

This project presents a low-cost, voice-controlled AI assistant built on the ESP32-S3 microcontroller. An I2S MEMS microphone (INMP441) captures speech and an on-device wake-word engine (ESP-SR) waits for the trigger phrase.

The recorded audio is streamed over WiFi to a cloud AI service, which converts speech to text, generates a reply with a large language model and returns synthesized speech. The reply is played through an I2S amplifier (MAX98357A) and a speaker.

The open-source XiaoZhi AI firmware forms the software base, and MCP tools allow the assistant to control appliances by voice. The result is a compact, affordable and extensible smart assistant.



Microcontroller with AI vector instructions

INTRODUCTION

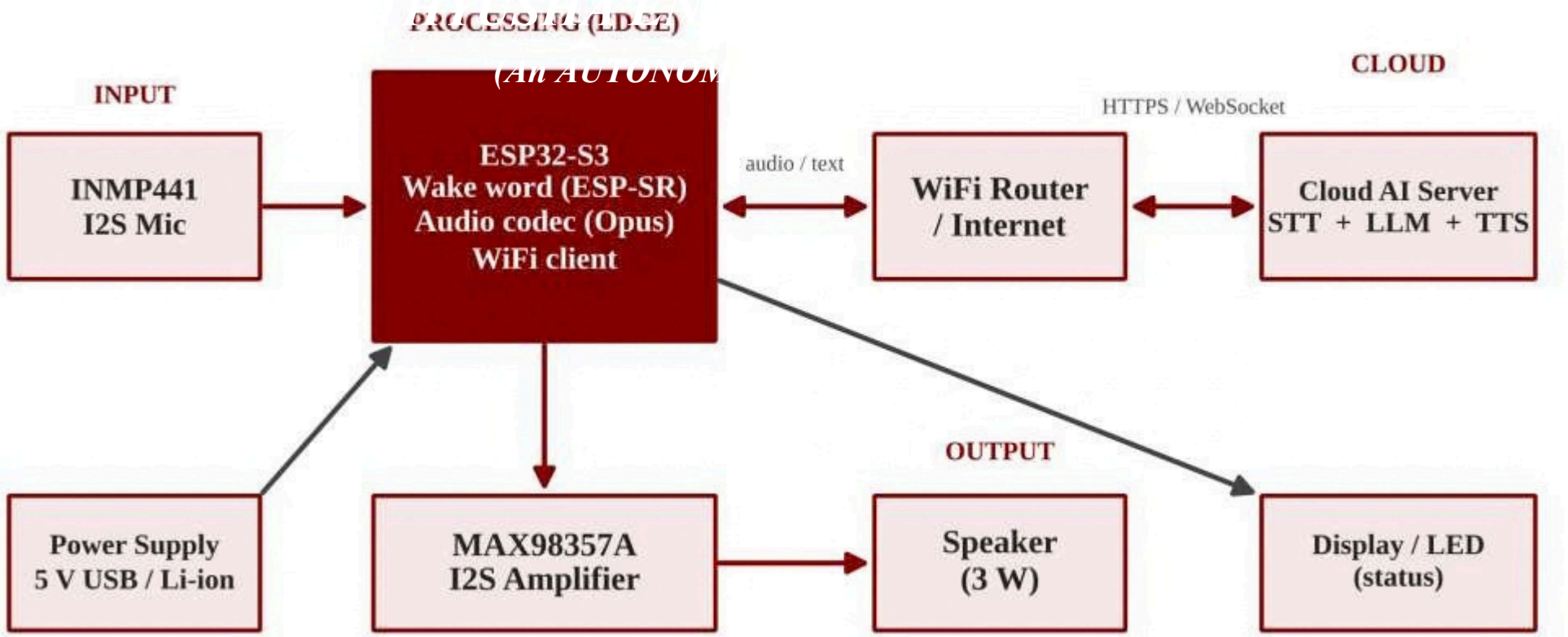
- Voice is the most natural wayto interactwith machines, but commercial assistantsare closed ecosystems withlimited customization.
- The ESP32-S3 is an inexpensive dual-core microcontroller with WiFi, Bluetooth 5 and AI vector instructions, well suited to audio front-ends.
- HeavyAI work (speech recognition, language model, speech synthesis) runs in the cloud, so the device itself stays small, cheap and low-power.
- Objective: wake on a spoken phrase, understand the request, answer intelligently and speak back in near real time using an open framework.



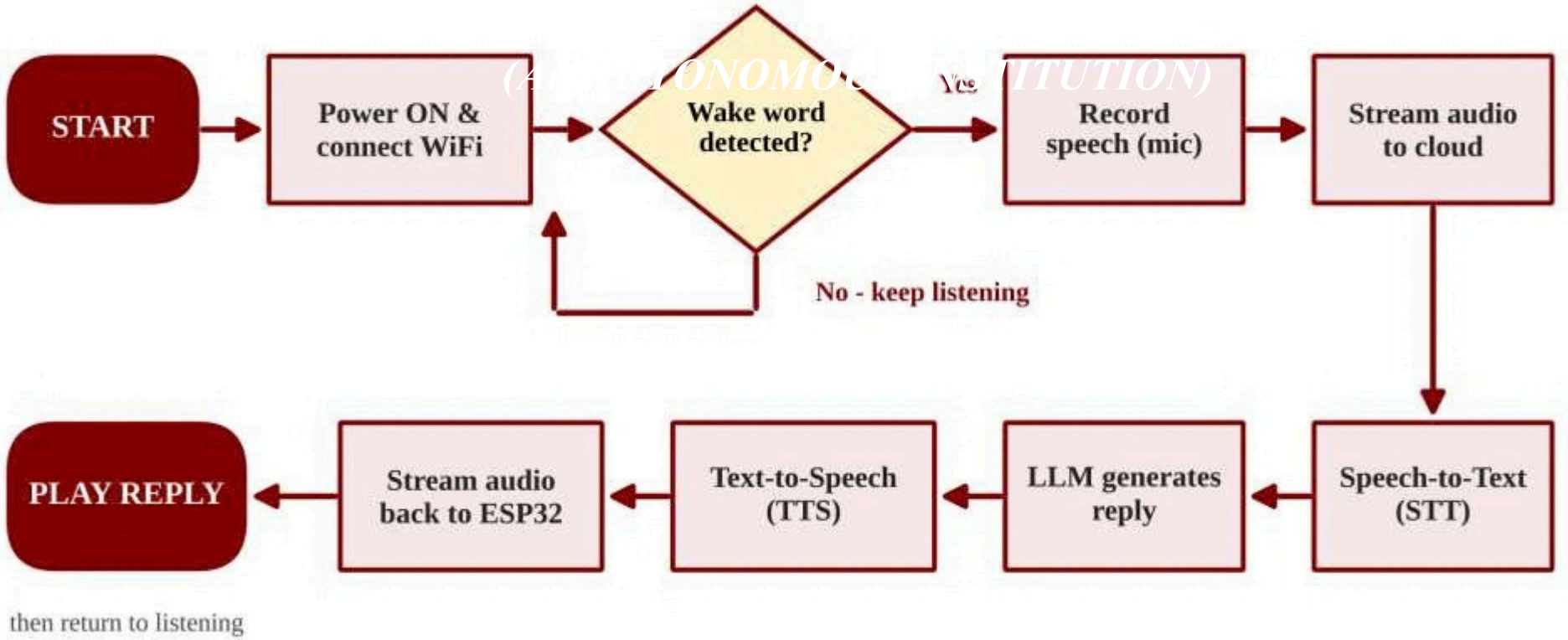
LITERATURE SURVEY

Reference	Work / finding	Relevance to thisproject
Hoy (2018)	"Alexa, Siri,Cortana,andMore"-overviewofcommercial voice assistants	Showstheir capabilities,limits and privacy concerns; motivates an open alternative
Zhang et al. (2017)	"Hello Edge" -keyword spotting with small neural networks on microcontrollers	Proves wake-word detection can run on-device with little memory
Warden (2018)	"Speech Commands" -open dataset for limited-vocabulary speech recognition	Standard data for training and testing keyword spotters
Radford et al. (2022)	Whisper -robust speech recognition trained on large-scale weak supervision	Accurate cloud speech-to-text across accents and noise
Brown et al. (2020)	"Language Models are Few-Shot Learners" (GPT-3)	Large language models give natural, context-aware replies
Espressif / XiaoZhi AI	ESP-SR wake-word framework and the open-source xiaozhi-esp32 firmware	Software base used to build the ESP32-S3 assistant

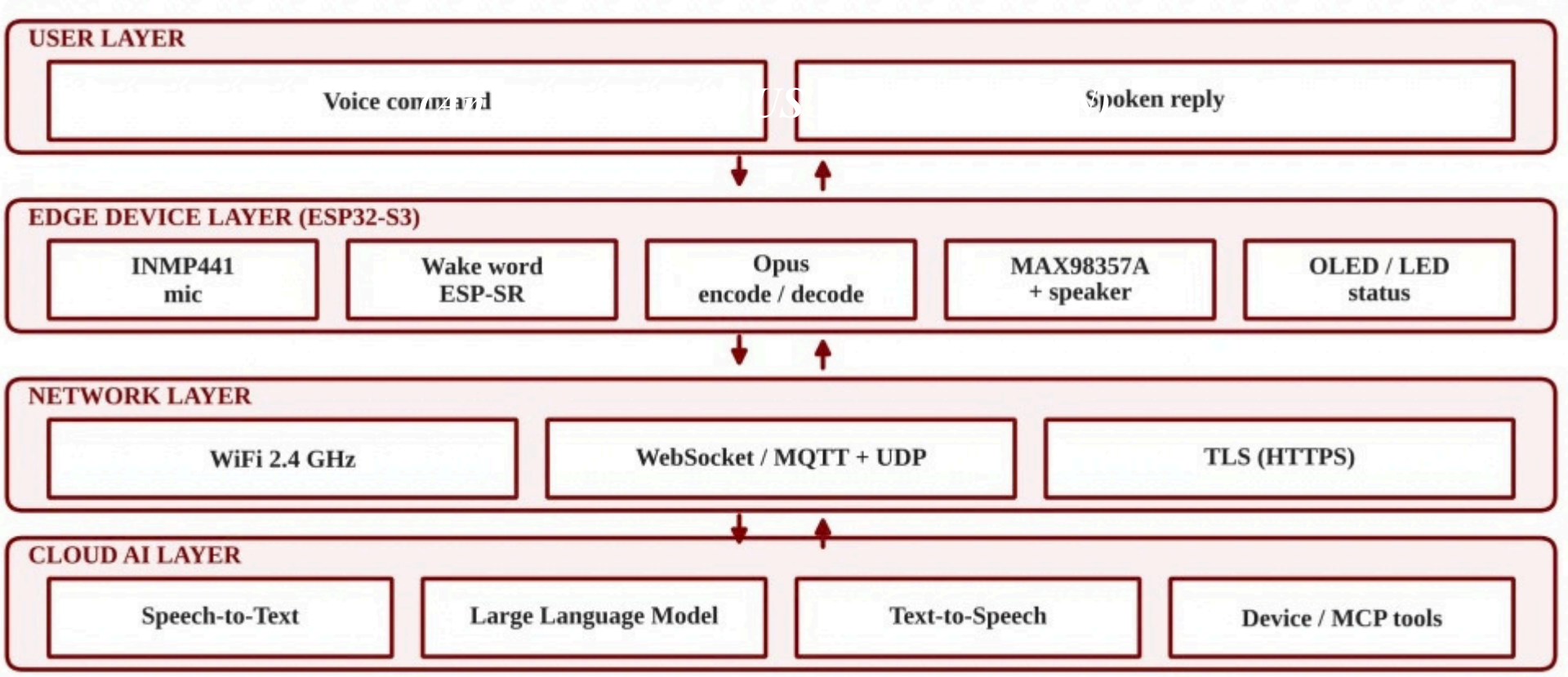
BLOCK DIAGRAM



FLOW CHART


START
Power ON & connect WiFi
Wake word detected?
Yes
No - keep listening
Record speech (mic)
Stream audio to cloud
Speech-to-Text (STT)
LLM generates reply
Text-to-Speech (TTS)
Stream audio back to ESP32
PLAY REPLY
then return to listening

SYSTEM ARCHITECTURE



HARDWARE COMPONENTS DESCRIPTION

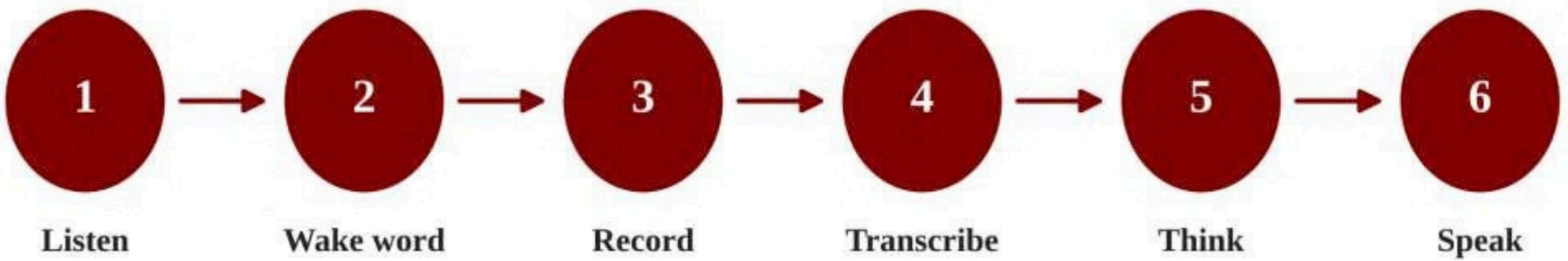
Component	Description
ESP32-S3 DevKit	Dual-core 240 MHz MCU with WiFi, BLE 5 and PSRAM; runs wake word and audio streaming
INMP441	I2S digital MEMS microphone, 24-bit, omnidirectional (3.3 V)
MAX98357A	I2S class-D amplifier that drives a 4-8 ohm speaker, up to about3W (5 V)
Speaker (3 W)	Converts the amplified audio into sound
OLED / LED	Optional status display: listening, thinking, speaking
Power supply	5 V USB or Li-ion cell with regulator



Connections: mic uses SCK, WS and SD; the amplifier uses BCLK, LRC and DIN.

Any free ESP32-S3 GPIOs can be assigned in the board configuration file; the mic runs on 3.3 V and the amplifier on 5V.

WORKINGPRINCIPLE



1 Listen: The ESP32-S3 continuously samples the INMP441 microphoneoverI2S.

2 Wake word: ESP-SR detects the trigger phrase on the device, so noaudioleavesuntilitiswoken.

3 Record: Speech is captured, compressed with Opus and streamed over WiFi.

4 Transcribe: Cloud speech-to-text converts the audio into text.

5 Think: The LLM understands the request and writes a reply, or calls an MCP tool such as a relay switch.

6 Speak: Text-to-speech audio streams back and plays through the MAX98357A and speaker.

SOFTWARE IMPLEMENTATION

- ***Firmware:*** XiaoZhi AI (xiaozhi-esp32), C/C++ on ESP-IDF 5.x with FreeRTOS tasks.
- ***Wake word:*** ESP-SR (WakeNet) with audio front-end for noise suppression and echo cancellation.
- ***Audio:*** I2S drivers for mic and amplifier; Opus codec for streaming.
- ***Network:*** WebSocket or MQTT + UDP, with OTA firmware updates.
- ***Cloud:*** STT, LLM and TTS on xiaozhi.me or a self-hosted xiaozhi-esp32-server.
- ***Extensions:*** MCP tools for GPIO, relays and smart-home control.

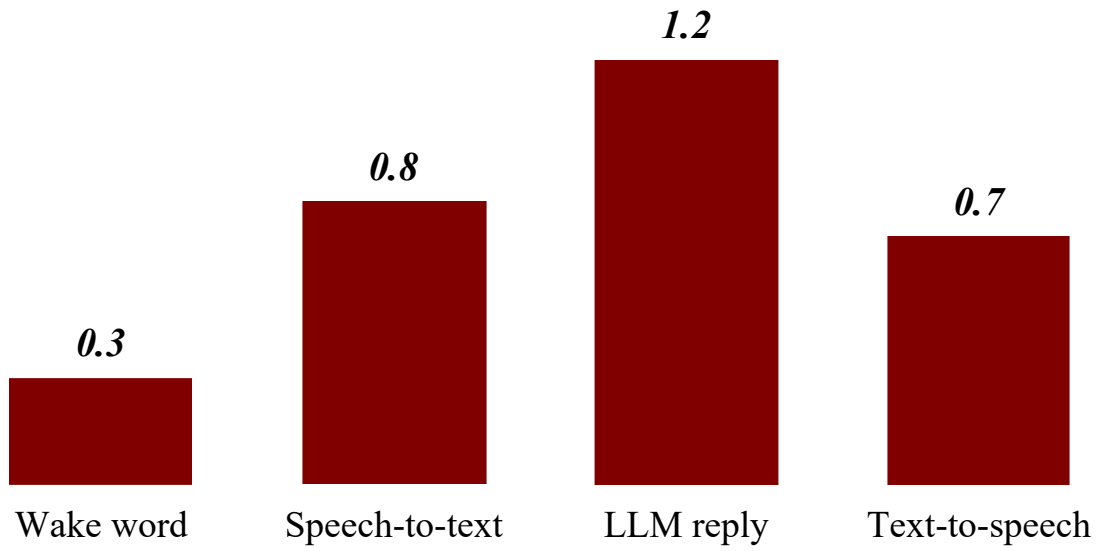
Build and flash

```
$ idf.py set-target esp32s3
$ idf.py menuconfig
$ idf.py build flash monitor
```

Then join the device hotspot, enter your WiFi details and bind the device with the code shown.

RESULTS AND DISCUSSION

Approximate response time per stage (s)



Indicative values; typical end-to-end reply is about 3 seconds.

- The device wakes on the trigger phrase and holds multi-turn conversationsover WiFi.
- Response time is dominated by network and LLM processing, not by the ESP32-S3.
- Recognition accuracy drops in noisy rooms; mic placement and the noise-suppression front-end help.
- An internet connection is required; a local server can be used for offline operation.

APPLICATIONS

Smart home

Voice control of lights, fans and appliances through relays and MCP tools.

Assistive technology

Hands-free access for visually or physically challenged users.

Education

Language practice and a hands-on AI learning kit for students.

In-vehicle assistant

Voice queries and controls without taking hands off the wheel.

Elder care

Reminders, companionship and simple voice help for older adults.

Industrial use

Hands-free commands and status queries on the shop floor.

ADVANTAGES

Lowcost
ESP32-S3 board and parts cost a few dollars.

Lowpower
Efficient MCU; wake word runs on-device.

Openandcustomizable
Open-source firmware and server; change voice, wake word or LLM.

Compact
WiFi, Bluetooth and audio support on one small board.

Naturalconversation
LLM gives flexible, multi-language, context-aware replies.

Easyto upgrade
OTA firmware updates and new MCP tools without new hardware.

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

- A working low-cost AI voice assistant was built on the ESP32-S3 using on-device wake-word detection and cloud speech, language and voice services.
- It shows that natural conversational AI is possible on a microcontroller costing a few dollars.
- The design is modular, open source and easy to extend.

Future enhancement

- Offline speech commands for faster, private operation.
- Custom wake word and more languages.
- Camera module for multimodal answers.
- Battery operation with sleep modes.
- Home Assistant integration and more MCP tools.

SDG/PO/PSO MAPPING WITH JUSTIFICATION

Mapping	Title	Justification
SDG 9	Industry, Innovation and Infrastructure	Innovative low-cost IoT and AI embedded system
SDG 4	Quality Education	Hands-on platform for learning AI, IoT and embedded design
SDG 10	Reduced Inequalities	Hands-free voice access helps people with disabilities
PO1	Engineering knowledge	Applies electronics, embedded and signal-processing concepts
PO3	Design and development of solutions	Designs the complete hardware and firmware system
PO5	Modern tool usage	Uses ESP-IDF, cloud AI services and an LLM
PO12	Life-long learning	Learns new AI, IoT and networking tools independently
PSO1	Embedded systems and IoT	Microcontroller-based product with sensors and actuators
PSO2	Communication systems	WiFi networking and real-time audio streaming

PROPOSED OUTCOME

- A working prototype that wakes on a spoken phrase and answersin naturallanguage.
- Voice control of home appliances through relays and MCP tools.
- Complete documentation: circuit diagram, firmware configuration and userguide.
- A low-cost bill of materials suitable for students and makers.
- A base platform that can be extended with new skills and sensors.



ESP32-S3 AI Voice Assistant